Product User Manual
Soil Sensor
(Temperature, Humidity, EC, Salt Content, Nitrogen, Phosphorus, Potassium)

Precautions

- Please follow the operating procedures and precautions in this manual during use.
- If the instrument malfunctions or is damaged during use, please contact the distributor; do not attempt to repair it yourself.
- Read this manual thoroughly before use.
- Do not insert the probe into stones or hard clods to avoid damaging the probe.
- Do not pull the cable directly when removing the sensor from the soil.
- Ensure the sensor probe is fully inserted into the soil/substrate to reduce operational errors and improve measurement accuracy.

Warning

- Incorrect wiring may damage this device and the connected instruments. Exceeding the maximum input power of the device will cause damage to the device.

Product Warranty

This product comes with a one-year warranty. Starting from the date of shipment, within twelve months, the company is responsible for free repair or replacement for faults caused by sensor quality issues (non-human damage). After the warranty period, only cost fees will be charged.

I.Introduction to the Soil Sensor

The soil sensor is a device capable of measuring various soil parameters, such as temperature, humidity, EC (Electrical Conductivity), salt content, nitrogen, phosphorus, and potassium. Through specific sensing elements, it converts physical, chemical, and biological information in the soil into electrical signals or digital signals for further processing and analysis. In modern agriculture, soil sensor technology is mainly used to monitor various soil parameters, providing data support for precision planting and intelligent management. With soil sensors, farmers can timely understand soil conditions, adjust management measures such as fertilization and irrigation, and improve the growth quality and yield of crops. Meanwhile, the application of this technology helps reduce production costs, protect the ecological environment, and realize the sustainable development of agriculture. The soil temperature part adopts **NTC** resistance collection with **12-bit ADC** conversion. The soil moisture part is based on the **FDR** (Frequency Domain Reflectometry) principle, which can directly and stably reflect the true moisture content of various soils by measuring the dielectric constant of the soil, and can measure the volumetric percentage of soil moisture. The soil electrical conductivity part uses a complementary **PWM** output as the signal excitation source and collects resistance values through a bridge circuit. Finally, temperature is used to compensate the electrical conductivity signal to **25**℃ before outputting it as a salt content signal.

II. Product Features

(1)The sensor adopts a needle-type design for easy insertion into the soil.

(2) High real-time data response and good interchangeability.

(3) Sealed with epoxy resin, achieving an **IP68** protection level.

(4) Less affected by soil type, suitable for a wide range of regions.

(5) Multiple electrical and magnetic isolation designs ensure product performance and normal operation.

(6) Standard **Modbus-RTU** protocol.

III.Application Scope

Suitable for water-saving agricultural irrigation, meteorological monitoring,

environmental monitoring, greenhouses, flowers and vegetables, grasslands and pastures, rapid soil testing, plant cultivation, scientific experiments, and other fields that require measuring soil temperature, humidity, electrical conductivity, soil pH, and soil nitrogen, phosphorus, and potassium.

IV. Technical Parameters

Parameter	Measuring Range	Unit	Accuracy	Resolution
Temperature	－40～80	℃	±0.5℃	0.1℃
Humidity	0～100	%(m^3/m^3)	0～50%（m^3/m^3）with in ±2%	0.1%
Electrical Conductivity (EC)	0～20000	μS/cm	±5%	1μS/cm
Salt Content	0～20000	μS/cm	±5%	1μS/cm
Nitrogen	0～1999	mg/kg（mg/L）	±3%（mg/kg）	1mg/kg（mg/L）
Phosphorus	0～1999	mg/kg（mg/L）	±3%（mg/kg）	1mg/kg（mg/L）
Potassium	0～1999	mg/kg（mg/L）	±3%（mg/kg）	1mg/kg（mg/L）

Output signal: A: Voltage signal (choose one of 0 ~ 2V, 0 ~ 5V, 0 ~ 10V)

B: 4 ~ 20mA (current loop)

C: RS485 (Standard Modbus-RTU protocol, default device address: 01)

Supply voltage

5 ~ 24V DC (when the output signal is 0 ~ 2V, RS485)

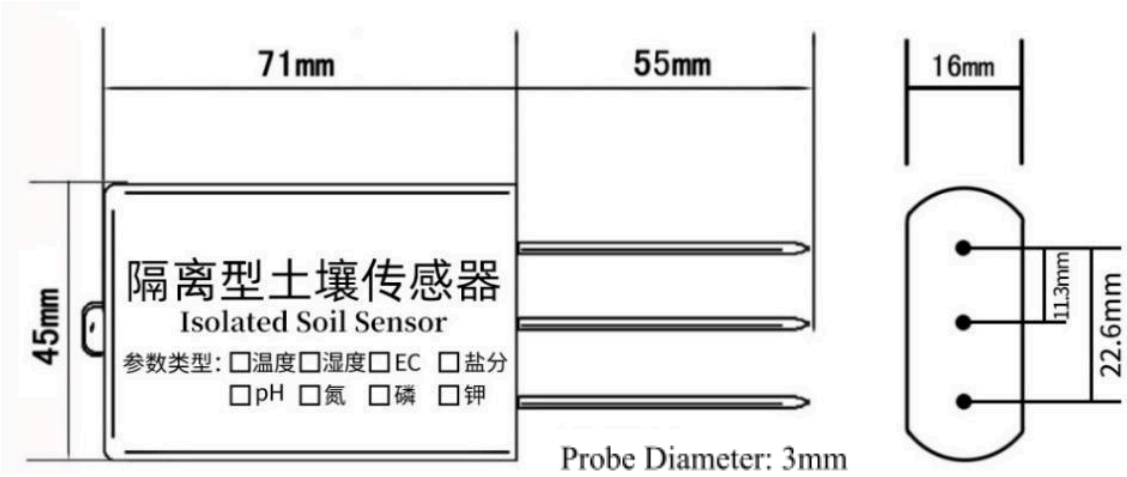
12 ~ 24V DC (when the output signal is 0 ~ 5V, 0 ~ 10V, 4 ~ 20mA)

Stabilization time: < 1 second

Response time: < 1 second

Electrical isolation: 1650V

Measurement area: A cylinder with a diameter of 7cm and a height of 7cm centered on the central probe



(Parameter Types: □Temperature □Humidity □EC □Salt Content □pH □Nitrogen □Phosphorus □Potassium)

VI.Wiring Instructions

The soil sensor can be connected to various data loggers, PLCs, etc., with four-core bare wires by default.

RS485	Category	Wire Color
Power Supply	Red	Positive Power Supply (VCC)(DC5V~24V)
	Black	Negative Power Supply (GND)
Communication	Yellow	485-A / Temperature(Analog)
	Green	485-B / Humidity(Analog)

Analog	Category	Wire Color
Power Supply	Red	Positive Power Supply (VCC)(DC12-24V)
	Black	Negative Power Supply (GND)
Communication	Yellow	Temperature
	Green	Humidity

Analog data impedance and conversion methods

Impedance requirements for current signals

Supply voltage	9V	12V	20V	24V

Maximum impedance	＜250Ω	＜400Ω	＜500Ω	＜900Ω

Simulation data conversion method

θv: Soil volumetric moisture content; T: Soil temperature;
V: The voltage value collected by the collector, unit: V;
A: The current value collected by the collector, unit: mA;

Output signal	Water conversion method	Temperature conversion method（－30～70℃）
0～2V DC	θv＝50*V	T＝50*V－30
0～5V DC	θv＝20*V	T＝20*V－30
0～10V DC	θv＝10*V	T＝10*V－30
4～20mA	θv＝6.25*A－25	T＝6.25*A－55

VII. Modbus-RTU Protocol Description

Modbus was invented by Modicon (now a brand of Schneider Electric) in 1979, and is the world's first bus protocol truly used in industrial fields. To better popularize and promote the distributed application of Modbus over Ethernet, Schneider Electric has transferred the ownership of the Modbus protocol to the IDA (Interface for Distributed Automation) organization, and established the Modbus-IDA organization, laying the foundation for the future development of Modbus. In China, Modbus has become a national standard (GB/T19582-2008).

7.1 Basic Communication Parameters

Encoding	8-bit Binary
Data Bits	8 bits
Parity Bit	None
Stop Bit	1 bit
Error Checking	CRC (Cyclic Redundancy Check)
Baud Rate	96000bit/s (factory default)

7.2 Definition of Data Frame Format

Adopting ModBus-RTU communication protocol, the format is as follows:

Initial Structure: ≥4 bytes of time

Address Code: 1 byte

Function Code: 1 byte

Data Area: N bytes

Error Checking: 16-bit CRC code

End Structure: ≥4 bytes of time

Address Code: The address of the sensor, unique in the communication network (factory default 0x01).

Function Code: Indicates the function of the command sent by the master.

Data Area: Specific communication data; note that 16-bit data is sent with the high byte first!

CRC Code: Two-byte check code.

Master Inquiry Frame Structure:

Address Code	Function Code	Starting Register Address	Register Length	CRC Low Byte	CRC High Byte
1 byte	1 byte	2 bytes	2 bytes	1 byte	1 byte

Slave Response Frame Structure:

Address Code	Function Code	Number of Valid Bytes	Data Area 1	Data Area 2	Data Area N	Check Code
1 byte	1 byte	1 byte	2 bytes	2 bytes	2 bytes	2 bytes

VIII. Protocol Parsing Instructions

8.1 Register Addresses

Register Address	Content	Operation	Definition Description
0000H~0007H	Valid Data Content	Read	Measurement data (16-bit unsigned integer)
0030 H	Device Address	Read/Write	1~253 (16-bit unsigned integer, factory default 1)
0031 H	Baud Rate	Read/Write	4800,9600,1920,5760,1152

※ Default Sensor Communication Protocol Description

Default baud rate: 9600 Baud, 8 data bits, no parity, 1 stop bit, station number 01, broadcast address FE.

8.2 Querying Data

Querying Data from a Sensor (Address: 1)

Master Send Request:

Address (8bit)	Function Code (8bit)	Starting Register Address High Byte (8bit)	Starting Register Address Low Byte (8bit)	Register Quantity High Byte (8bit)	Register Quantity Low Byte (8bit)	CRC Low Byte (8bit)	CRC High Byte (8bit)
0x01	0x03	0x00	0x00	0x00	0x07	0x04	0x08

Sensor Normal Response:

Address	0X01	
Function Code	0X03	
Data Length	0X0E	
Register 0 Data High	0XFF	Soil Temperature: -3.5'C(Negative numbers are represented in two's complement)
Register 0 Data Low	0XDD	
Register 1 Data High	0X01	Soil Humidity: 35.6%
Register 1 Data Low	0X64	
Register 2 Data High	0X04	Soil Electrical Conductivity:1234μS/cm
Register 2 Data Low	0XD2	
Register 3 Data High	0X05	Soil Salt Content: 1356μS/cm
Register 3 Data Low	0X4C	
Register 4 Data High	0X00	Soil Nitrogen: 135mg/kg
Register 4 Data Low	0X87	
Register 5 Data High	0X00	Soil Phosphorus: 138mg/kg
Register 5 Data Low	0X8A	
Register 6 Data High	0X00	Soil Potassium: 142mg/kg
Register 6 Data Low	0X8E	
CRC16 LOW	0XB7	
CRC16 High	0XDF	

Data Representation Method:

A. Soil Temperature: Convert the returned data to a decimal number and divide by 10. For negative numbers, use two's complement algorithm: subtract the data from 0XFFFF, add 1, convert to a decimal number, divide by 10, then take the negative.

B. Soil Humidity: Convert the data to a decimal number and divide by 10.

C. Soil Electrical Conductivity: Convert the data to a decimal number.

D. Soil Salt Content: Convert the data to a decimal number.

E. Soil Nitrogen: Convert the data to a decimal number.

F. Soil Phosphorus: Convert the data to a decimal number.

G. Soil Potassium: Convert the data to a decimal number.

8.2.1、Querying the Current Device Address

Master Send Request:

Address (8bit)	Function Code (8bit)	Starting Register Address High Byte (8bit)	Starting Register Address Low Byte (8bit)	Register Quantity High Byte (8bit)	Register Quantity Low Byte (8bit)	CRC Low Byte (8bit)	CRC High Byte (8bit)
0xFE	0x03	0x00	0x30	0x00	0x01	0x90	0x0A

Sensor Normal Response:

Address (8bit)	Function Code (8bit)	Number of Bytes (8bit)	Data High Byte (8bit)	Data Low Byte (8bit)	CRC Low Byte (8bit)	CRC High Byte (8bit)
0x01	0x03	0x02	0x00	0x01	0x79	0x84

Current Device Address: 0x01 (Data Low Byte)

8.2.2 Modifying the Current Device Address

Example: Changing the address of a sensor from 1 to 2

Master Send Request:

Address (8bit)	Function Code (8bit)	Starting Register Address High Byte (8bit)	Starting Register Address Low Byte (8bit)	Register Quantity High Byte (8bit)	Register Quantity Low Byte (8bit)	CRC Low Byte (8bit)	CRC High Byte (8bit)
0x01	0x06	0x00	0x30	0x00	0x02	0x08	0x04

Sensor Normal Response:

Address (8bit)	Function Code (8bit)	Starting Register Address High Byte (8bit)	Starting Register Address Low Byte (8bit)	Register Quantity High Byte (8bit)	Register Quantity Low Byte (8bit)	CRC Low Byte (8bit)	CRC High Byte (8bit)
0x01	0x06	0x00	0x30	0x00	0x02	0x08	0x04

If the sensor receives the command correctly, it will return the data via the same path.

8.3 Modifying the Baud Rate

Example: Changing the baud rate from 9600 to 19200 (Master → Slave)

Master Send Request:

Address (8bit)	Function Code (8bit)	Starting Register Address High Byte (8bit)	Starting Register Address Low Byte (8bit)	Register Quantity High Byte (8bit)	Register Quantity Low Byte (8bit)	CRC Low Byte (8bit)	CRC High Byte (8bit)
0x01	0x06	0x00	0x31	0x19	0x20	0xD2	0x4D

Sensor Normal Response:

Address (8bit)	Function Code (8bit)	Starting Register Address	Starting Register Address	Register Quantity High Byte	Register Quantity Low Byte	CRC Low Byte	CRC High Byte

		High Byte (8bit)	Low Byte (8bit)	(8bit)	(8bit)	(8bit)	(8bit)
0x01	0x06	0x00	0x31	0x19	0x20	0xD2	0x4D

If the sensor receives the command correctly, it will return the data via the same path. In the example, writing data 1920 to register 00 31 indicates 19200.